

h2kweb

Enterprise Migration Dashboard

Browser-Based VM Migration, Monitoring & Management

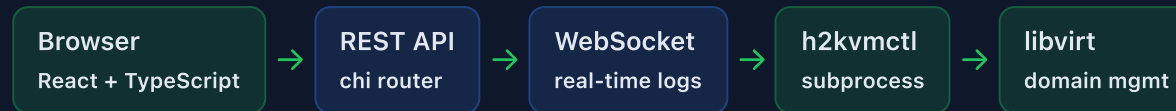
[Go Backend](#) | [React Frontend](#) | [PAM Auth](#) | [WebSocket](#) | [Libvirt](#) | [Proprietary \(HyperSDK\)](#)

github.com/ssahani/hyper2kvm

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1. Architecture

Full-stack web platform with Go backend, React/TypeScript frontend, and real-time WebSocket communication.



Go Backend (h2kweb)

- chi router with middleware (CORS, auth, logging)
- PAM authentication — system-level credentials
- HttpOnly cookies with SameSite=Strict
- 32-byte random session tokens
- WebSocket hub for real-time log streaming
- libvirt integration via go-libvirt bindings

React/TypeScript Frontend

- Vite build system with hot module replacement
- TailwindCSS + shadcn/ui component library
- Dark and light theme support
- Responsive layout (desktop + tablet)
- WebSocket client for live migration logs
- noVNC integration for VM console access

API Endpoints

- `POST /api/auth/login`
- `GET /api/dashboard`
- `GET /api/vms`
- `POST /api/migrate`
- `GET /api/jobs/:id/logs`
- `WS /api/ws`

Security

- PAM auth (same as SSH credentials)
- Session-based with secure cookies
- CSRF protection via SameSite
- No default passwords
- Optional API key for automation

Integration

- libvirt: domain CRUD, screenshots
- h2kvmctl: migration subprocess
- govc: VMware vSphere discovery
- KubeVirt: Kubernetes VM deploy
- systemd: service management

Go chi router

React + TypeScript

PAM authentication

WebSocket real-time

libvirt integration

Dark/Light themes

2. Dashboard Overview

Single-pane view of migration pipeline, host resources, readiness checks, and recent activity.

Migration Pipeline

- Active migrations with real-time progress bars
- Queued, running, completed, and failed job counts
- Per-job elapsed time and ETA
- Click-through to full job logs
- Cancel in-progress migrations

Host Resources

- CPU: model, cores, current load average
- Memory: total, used, available with percentage
- Disk: per-mount usage with warnings at 90%+
- QEMU version and KVM status
- libvirt connection state

Readiness Checks (19 Items)

- KVM hardware acceleration available
- QEMU binary found and executable
- libvirt daemon running and accessible
- qemu-nbd and nbd kernel module loaded
- virtio-win ISO present and valid
- OVMF firmware files found
- hivex Python bindings available
- Sufficient disk space for conversions
- Network bridge or NAT configured

Disk Images Inventory

- Scan configured directories for VMDK, OVA, QCOW2
- File size, format, and modification date
- Quick-action: migrate selected image
- Delete or move completed conversions
- Storage usage summary per directory

All 19 readiness checks must pass before migrations can start. The dashboard shows a clear green/amber/red status for each check, with remediation hints for any failures.

19 readiness checks

Real-time CPU/RAM/disk

Pipeline status

Disk image inventory

Recent activity feed

3. VM Management

Full libvirt domain management with live screenshots, VNC console, and guest agent integration.

VM Information

- OS type detection (Linux/Windows)
- Disk bus and format (virtio/sata, qcow2/raw)
- vCPU count and memory allocation
- Network interfaces with MAC addresses
- Firmware type (UEFI/BIOS)
- Autostart and persistent status

Guest Agent Status

- Agent ping: connected or disconnected
- Guest OS version from agent
- IP addresses reported by agent
- Filesystem usage from agent
- Hostname and timezone
- User sessions active

Console & Screenshots

- noVNC in-browser console (no client needed)
- Live screenshots via libvirt API
- Screenshot refresh on demand
- Full-screen console mode
- Clipboard sharing (with agent)
- Ctrl+Alt+Del send button

VM Actions

Action	API	Description
Start	POST /api/vms/:id/start	Boot the VM, wait for guest agent ping confirmation
Shutdown	POST /api/vms/:id/shutdown	ACPI shutdown signal, graceful guest OS shutdown
Force Off	POST /api/vms/:id/destroy	Immediate power-off (like pulling the plug)
Reboot	POST /api/vms/:id/reboot	ACPI reboot via guest agent or reset
Delete	DELETE /api/vms/:id	Undefine domain, optionally remove disk images
Console	WS /api/vms/:id/console	WebSocket proxy to VNC for noVNC client

noVNC console

Live screenshots

Guest agent integration

CPU/memory stats

OS type detection

Disk bus/format info

4. Migration Workflow

Browser-based file upload, 3-step migration wizard, and live log streaming with auto-scroll.

Upload / Select
drag-drop or browse



Configure
3-step wizard



Convert
h2kvmctl subprocess



Deploy
libvirt + boot

File Upload

- Drag-and-drop zone for VMDK, OVA, VHD files
- Up to 50 GB upload size supported
- Real-time upload progress bar with percentage
- Cancel upload mid-transfer
- Chunked upload for reliability over slow connections
- Server-side directory browse for local files

3-Step Wizard

- **Step 1:** Source — file path, format auto-detect
- **Step 2:** Options — output dir, flatten, compress, fixes
- **Step 3:** Deploy — libvirt XML, virsh define, boot test
- YAML preview panel shows generated config
- All h2kvmctl options exposed in advanced section
- Save and reuse migration templates

Live Logs

- WebSocket streaming of h2kvmctl stdout/stderr
- Auto-scroll follows latest output
- Progress bar with percentage, speed, and ETA
- Color-coded: info (blue), warning (amber), error (red)
- Search within log output
- Download full log as text file

Report Export

- Migration summary report (JSON and text)
- Source and destination details
- Duration, disk sizes, conversion ratio
- Fixes applied (drivers, initramfs, GRUB)
- Deploy result: domain name, IP address
- Shareable link to migration result page

Drag-drop upload (50GB)

3-step wizard

WebSocket live logs

Progress + ETA

YAML preview

Report export

5. Network & Infrastructure

Network topology visualization, libvirt domain management, and KubeVirt integration.

Network Topology

- Visual map of libvirt networks
- VMs grouped by network (default, bridge, NAT)
- IP addresses per VM (from guest agent)
- MAC addresses and interface names
- Network type and DHCP range display
- Bridge device mapping

Libvirt Management

- Domain list with state and resource usage
- Storage pool management (dir, LVM, NFS)
- Network management (create, edit, delete)
- Snapshot management per VM
- Domain XML editor with syntax highlighting
- Bulk operations on selected VMs

KubeVirt Integration

- Deploy migrated VMs to Kubernetes clusters
- VirtualMachine manifest generation
- PVC-backed disk configuration
- Network attachment definitions
- kubectl apply from dashboard
- K3s single-node support

Provider Integration

Provider	Protocol	Discovery	Migration
VMware vSphere	govmomi / govc	Datacenter, cluster, VM listing	OVA export + convert + deploy
Microsoft Azure	az CLI	Subscription, resource group, VM listing	VHD download + convert + deploy
AWS EC2	aws CLI	Region, VPC, instance listing	AMI export + convert + deploy
Local File	filesystem	Directory browse, file picker	Direct convert + deploy

Network topology view

3 cloud providers

Libvirt domain management

KubeVirt deploy

Storage pool management

6. Deployment

One-command remote deployment, systemd service, watch directory daemon, and theme support.

deploy-remote.sh

```
# One-command setup on any Linux host
curl -sL https://raw.githubusercontent.com/ssahani/hyper2kvm/main/scripts/
  deploy-remote.sh | bash -s -- \
    --host 10.0.1.50 \
    --user root
```

- rsync project files to remote host
- pip install with dependency resolution
- hivex source rebuild for Python 3.12
- virtio-win ISO auto-download
- systemd service installation

Daemon Service

```
# Watch directory for auto-conversion
[Service]
ExecStart=/usr/local/sbin/h2kvmctl \
  --daemon \
  --watch-dir /var/lib/hyper2kvm/incoming \
  --output-dir /var/lib/hyper2kvm/converted \
  --auto-deploy
```

- inotify watch on incoming directory
- Auto-convert dropped VMDK/OVA/VHD files
- Auto-deploy to libvirt after conversion
- Configurable concurrency (default: 1)

h2kweb systemd

- Unit: `h2kweb.service`
- Default port: 5070
- Auto-restart on failure
- Journal logging integration
- Socket activation support

Dark / Light Theme

- System preference auto-detection
- Manual toggle in header
- Persistent via localStorage
- All components themed
- Chart colors adapt

Configuration

- `--binary` : h2kvmctl path (auto-detect)
- `--addr` : listen address (:5070)
- `--api-key` : optional API auth
- `--static-dir` : frontend assets
- `--tls-cert/key` : HTTPS

Enterprise-Ready Web Dashboard

One-command deployment. Browser-based migration. Real-time monitoring. Zero desktop software required.

PAM auth + WebSocket logs + libvirt integration + KubeVirt deploy = complete migration platform.